$$\begin{aligned}
\text{1Q HPR: } r_1 &= (\text{USD}6{,}000{,}000 - \text{USD}5{,}000{,}000) / \text{USD}5{,}000{,}000 = 0.20\\
\text{2Q HPR: } r_2 &= (\text{USD}5{,}775{,}000 - \text{USD}5{,}500{,}000) / \text{USD}5{,}500{,}000 = 0.05\\
\text{3Q HPR: } r_3 &= (\text{USD}6{,}720{,}000 - \text{USD}6{,}000{,}000) / \text{USD}6{,}000{,}000 = 0.12\\
\text{4Q HPR: } r_4 &= (\text{USD}5{,}508{,}000 - \text{USD}6{,}120{,}000) / \text{USD}6{,}120{,}000 = -0.10
\end{aligned}$$

$$R_{\text{TW}} = (1 + r_1)(1 + r_2)(1 + r_3)(1 + r_4) - 1,$$

$$R_{\text{TW}} = (1.20)(1.05)(1.12)(0.90) - 1 = 0.2701 \text{ or } 27.01\%.$$

2. Calculate the time-weighted rate of return for the Super Trust account.

Solution:

The account managed by Super Trust has a time-weighted rate of return of 26.02 percent, calculated as follows:

$$\begin{aligned}
\text{1Q HPR: } r_1 &= (USD13,200,000 - USD12,000,000) / USD12,000,000 = 0.10\\
\text{2Q HPR: } r_2 &= (USD12,240,000 - USD12,000,000) / USD12,000,000 = 0.02\\
\text{3Q HPR: } r_3 &= (USD5,659,200 - USD5,240,000) / USD5,240,000 = 0.08\\
\text{4Q HPR: } r_4 &= (USD5,469,568 - USD5,259,200) / USD5,259,200 = 0.04
\end{aligned}$$

$$R_{\text{TW}} = (1 + r_1)(1 + r_2)(1 + r_3)(1 + r_4) - 1,$$

$$R_{\text{TW}} = (1.10)(1.02)(1.08)(1.04) - 1 = 0.2602 \text{ or } 26.02\%.$$

The in-house portfolio's time-weighted rate of return is higher than the Super Trust portfolio's by 99 basis points. Note that 27.01 percent and 26.02 percent might be rounded to 27 percent and 26 percent, respectively. The impact of the rounding the performance difference (100 bp vs. 99 bp) may seem as trivial, yet it's impact on a large portfolio may be substantive.

Having worked through this exercise, we are ready to look at a more detailed case.

EXAMPLE 10

Time-Weighted and Money-Weighted Rates of Return Side by Side

Your task is to compute the investment performance of the Walbright Fund for the most recent year. The facts are as follows:

- On 1 January, the Walbright Fund had a market value of USD100 million.
- During the period 1 January to 30 April, the stocks in the fund generated a capital gain of USD10 million.
- On 1 May, the stocks in the fund paid a total dividend of USD2 million. All dividends were reinvested in additional shares.
- Because the fund's performance had been exceptional, institutions invested an additional USD20 million in Walbright on 1 May, raising assets under management to USD132 million (USD100 + USD10 + USD2 + USD20).
- On 31 December, Walbright received total dividends of USD2.64 million. The fund's market value on 31 December, not including the USD2.64 million in dividends, was USD140 million.
- The fund made no other interim cash payments during the year.